

RESEARCH INTERESTS

Random matrix theory, high-dimensional probability, AI for mathematics, theoretical computer science, and mathematical foundations of data science.

PUBLICATIONS AND PREPRINTS

- On the Subgaussianity of Quantized Linear Maps: An AI-Assisted Note2026  
Guangyi Zou and Roman Vershynin; arXiv:2605.27563; 3rd AI for Math Workshop at ICML 2026
- Edge Universality for Inhomogeneous Random Matrices II: Markov Chain Comparison and Critical Statistics2026  
Dang-Zheng Liu and Guangyi Zou; arXiv:2604.20215; preprint
- Edge Universality for Inhomogeneous Random Matrices2025  
Dang-Zheng Liu and Guangyi Zou; arXiv:2508.17838; submitted
- Revisit the Partial Coloring Method: Prefix Spencer and Sampling2025  
Dongrun Cai, Xue Chen, Wenxuan Shu, Haoyu Wang, and Guangyi Zou; arXiv:2408.13756; COCOON 2025
- Outliers for Deformed Inhomogeneous Random Matrices2024  
Ruohan Geng, Dang-Zheng Liu, and Guangyi Zou; arXiv:2407.12182; submitted
- Edge Statistics for Random Band Matrices2024  
Dang-Zheng Liu and Guangyi Zou; arXiv:2401.00492; submitted

EDUCATION

|       |                                               |                                |                     |
|-------|-----------------------------------------------|--------------------------------|---------------------|
| Ph.D. | University of California, Irvine              | Advisor: Prof. Roman Vershynin | Sep 2025 – Present  |
| M.Sc. | University of Science and Technology of China | Advisor: Prof. Dang-Zheng Liu  | Sep 2022 – Jun 2024 |
| B.Sc. | University of Science and Technology of China |                                | Sep 2018 – Jun 2022 |

RESEARCH EXPERIENCE

- Research Assistant, University of Science and Technology of ChinaMar 2025 – Jul 2025  
Supervisor: Prof. Dang-Zheng Liu
- Research Assistant, Tsinghua UniversitySep 2024 – Jan 2025  
Supervisor: Prof. Fan Yang
- Academic Visitor, Peking UniversityJan 2022 – May 2022  
Host: Prof. Xinyi Li

AWARDS

|                                                                                         |            |
|-----------------------------------------------------------------------------------------|------------|
| Euler Outstanding Promise Award                                                         | 2026       |
| Anhui Province Outstanding Graduate (top 4%)                                            | 2024       |
| Graduate Scholarship (2×10,000 RMB)                                                     | 2022, 2023 |
| One Silver Prize and Three Bronze Prizes, S.-T. Yau College Student Mathematics Contest | 2020, 2021 |
| Yang Yuanqing Education Foundation Scholarship (top 1%)                                 | 2020       |

LEADERSHIP AND TEACHING

- Teaching Assistant, Probability TheoryFall 2021; Spring 2022, 2023, 2024  
University of Science and Technology of China
- President, Student Algorithm ClubJun 2020 – Jun 2021  
University of Science and Technology of China

TECHNICAL SKILLS

Programming: Python, C++      Typesetting: L<sup>A</sup>T<sub>E</sub>X